

Ferdaous Masmoudi

PhD in Electrical Engineering
Energy Systems Modeling,
Control & Optimization

Route Gremda
3062 Sfax, Tunisia
French residence permit
☎ +216 23 605 813

✉ ferdaous.masmoudi@gmail.com
🌐 ferdaousmasmoudi.github.io
ID 0000-0001-8457-7162
R[®] Ferdaous-Masmoudi
in [ferdaous-masmoudi-2a95b64a](#)
🐙 [ferdaousmasmoudi](#)



Research Profile

PhD in Electrical Engineering and Renewable Energy Systems with experience in the end-to-end development of intelligent energy platforms, covering nonlinear mathematical modeling, parameter identification, optimization strategies, embedded implementation, and experimental validation.

Research activities include multi-level mathematical modeling of photovoltaic and electrochemical energy systems, data-driven parameter identification (least squares, ARX-based approaches), and optimization-based tuning under real environmental conditions. State estimation techniques, including Extended Kalman Filtering (EKF), were implemented for real-time monitoring and control-oriented system supervision.

Strong experience in the integration of sensors, real-time data acquisition chains, signal processing, and control algorithms within embedded architectures, complemented by MATLAB/Simulink-based simulation, power electronics implementation, and cloud-enabled supervision platforms for real-world system deployment.

Education

- 2016 **PhD in Electrical Engineering**, *National Engineering School of Sfax (ENIS), University of Sfax, Tunisia*, Thesis: *Optimization of Standalone Photovoltaic Systems*.
- 2012 **MSc in Electrical Conversion and Renewable Energy (CEER)**, *ENIS, University of Sfax, Tunisia*, Research: *Modeling and MPPT Control of a PV Conversion System*.
- 2011 **Engineering Degree in Electrical Engineering (Electro-Technics)**, *ENIS, Sfax, Tunisia*
- 2008 **Preparatory Cycle for Engineering Studies (Mathematics–Physics)**, *IPEIM El Manar, Tunis, Tunisia*
- 2006 **Baccalaureate in Mathematics, with Honors**, *Lycée Pilote Ariana (LPA), Ariana, Tunisia*

Doctoral Research Contributions

Nonlinear Modeling and Identification	Development of nonlinear mathematical models of photovoltaic generators (single-, double-, and multi-diode configurations, panel-level modeling) and electrochemical storage systems, including analytical formulation, state-space representation, and data-driven parameter identification and estimation under real operating conditions.
Optimization and Control	Formulation and implementation of optimization-based control strategies for photovoltaic energy systems, including MPPT algorithms, dual-axis solar tracking mechanisms, and boost-converter power regulation, with performance validation under dynamic irradiance and temperature variations.
Embedded Implementation and Experimental Validation	MATLAB/Simulink-based simulation and model validation, followed by implementation of embedded control architectures (C++/Python) on SoC and microcontroller platforms for real-time operation. Design and realization of a boost converter power stage and dual-axis solar tracking mechanism with motor drivers integration. Development of real-time multi-sensor acquisition systems (voltage, current, irradiance, temperature, GPS, RTC, IMU) including signal conditioning, filtering, and experimental dataset validation. Deployment of cloud-enabled supervision and server-side data logging for remote monitoring and system performance analysis.

Applied Research & Engineering Experience

- 2016–Present **Co-founder and R&D Lead, NOVEL-TI, Sfax, Tunisia**
- Directed applied research and system development in intelligent energy systems, embedded architectures, and AI-enabled IoT platforms.
 - Designed and supervised end-to-end development of hardware–software systems, including mobile robotic platforms, photovoltaic-powered systems, multi-sensor acquisition architectures, and cloud-connected supervision dashboards.
 - Led development of AgriTech solutions integrating LoRa/WiFi connectivity, RFID/NFC logging, and blockchain-based traceability infrastructures (Hyperledger Fabric) for compliance-driven industrial environments.
 - Managed multidisciplinary technical teams and coordinated international collaborations with European industrial partners.

Academic Internships & Early Industrial Training

- 2013–2014 **Research Internship – Battery Observation and Management System (BOMS), LSIS Laboratory, Polytech Marseille, Marseille, France**
- Modeling and embedded implementation of battery monitoring and energy management algorithms, with laboratory validation using MATLAB and microcontroller-based architectures.
- 2012 **Research Internship – Residential PV Integration, TU Chemnitz, Chemnitz, Germany**
- Simulation and component-level optimization of photovoltaic system integration in residential environments.
- 2009–2011 **Industrial Training – Energy and Power Systems (Tunisia):** STEG (National Dispatching), OACA – Carthage Airport, SNDP AGIL. Exposure to grid operation, medium-voltage system design, switchgear manufacturing, and grid-connected photovoltaic system sizing.

Teaching Experience

- 2016–Present Contract-based Lecturer and Research Collaborator. Delivered lectures and laboratory sessions in instrumentation, mechatronics, embedded systems, and open-source development across multiple higher education institutions.
- Teaching Institutions:** Faculty of Sciences of Sfax (FSS); Higher Institute of Industrial Management (ISGI); Private Institute of Applied Sciences (IPSAS).
- Courses and Labs:** Computer-Based Instrumentation; Open-Source Software Development; Maintenance of Mechatronic Systems; Mechatronics.
- Participation in IEEE SSD'18 and technical workshops on IoT, STM32, Arduino, and smart systems.

Technical Skills

Programming & Build Systems	C++, Python, MATLAB, Bash scripting, GNU Make	Embedded Platforms	STM32, ESP32, Arduino, Jetson, Raspberry Pi, Odroid, Motor Driver
Simulation & Instrumentation	MATLAB/Simulink, LabVIEW, Proteus (ISIS/ARES)	PCB Design& Hardware	Altium, KiCad, EasyEDA, firmware development, bootloader configuration
Communication & IoT	MQTT, LoRa, Zigbee, Wi-Fi, BLE, GSM, Modbus, I2C, SPI, UART	Robotics Middleware	ROS/ROS2, Gazebo, RViz
Mobile & Web Development	Android (Kotlin), REST APIs, Firebase, QR/Camera integration	Cloud & Backend	Real-time data logging, server-side supervision, cloud synchronization
AI & Vision	OpenCV (C++/Python), YOLO, TensorFlow Lite, ONNX, RL(basic)	Platforms & Tools	Linux, Windows, Git

Languages

Arabic (Native) · French (Fluent) · English (Professional proficiency) · German (Basic knowledge)

Selected Publications

- Peer-Reviewed Journals
- Masmoudi, F., "Identification of internal parameters of a mono-crystalline photovoltaic cell model with experimental validation," *International Journal of Renewable Energy Research (IJRER)*, 2014 (Indexed in Scopus).
 - Jarraya, I., Masmoudi, F., et al., "An online state-of-charge estimation for lithium-ion and supercapacitor in hybrid electric drive vehicle," *Journal of Energy Storage*, 2019 (Elsevier, Scopus indexed).
 - Loukil, J., Masmoudi, F., Derbel, N., "A real-time estimator for model parameters and state-of-charge of lead-acid batteries in photovoltaic applications," *Journal of Energy Storage*, 2021 (Elsevier, Scopus indexed).
- International Conferences (IEEE Xplore Indexed)
- SSD 2019 – MPPT control strategies for photovoltaic applications: algorithms and comparative analysis.
 - SSD 2018 – "Modeling and parameters estimation for lithium-ion cells in electric drive vehicle,"
 - SSD 2018 – "Performance analysis of Luo-converter for photovoltaic application,"
 - SSD 2017 – "Design and comparison of quadratic boost and double cascade boost converters with boost converter,"
 - SSD 2017 – "Comparative study of fundamental variable input converters used for PV conversion systems,"
 - SSD 2017 – "State-of-charge estimation of lead-acid battery using a Kalman filter,"
 - SM2C 2017 – "Comparative study of the performances of the DC/DC Luo-converter in photovoltaic applications,"
 - IREC 2016 – "Modeling and simulation of conventional DC–DC converters dedicated to photovoltaic applications,"
 - SSD 2016 – "Single and double diode models for conventional mono-crystalline solar cell with extraction of internal parameters,"
 - SSD 2016 – "Modeling of internal parameters of a lead-acid battery with experimental validation,"
 - STA 2015 – "Design and realisation of a photovoltaic system controlled by an MPPT algorithm,"
 - SAAEI 2014 – "Simple models design for one mono-crystalline photovoltaic cell by identification of internal parameters." Maroc (Tanger).
 - EVER 2012 – "Oil service station PV-based power supply: a challenging Tunisian national program," (Monaco).
- Book Chapters (Springer)
- Loukil, J., Masmoudi, F., Derbel, N., "Third order model and identification of lead-acid batteries using meta-heuristic algorithms and experimental measurements," Book Chapter, 2020.
 - Boujelben, N., Masmoudi, F., Djemel, M., Derbel, N., "Modeling and comparison of boost converter with cascaded boost converters," *Green Energy and Technology*, Book Chapter, 2019.

Online Research Portfolio

Scan to access the full thesis, publications, and research materials.



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FLOTTANT Recruitment Committee

February 23, 2026

Université de Sfax

École Nationale d'Ingénieurs de Sfax (ENIS)

Sfax, Tunisia

Dear Members of the Recruitment Committee,

I am writing to express my strong interest in the postdoctoral position within the FLOTTANT project. I hold a PhD in Electrical Engineering with specialization in renewable energy systems, nonlinear mathematical modeling, optimization-based control, and state estimation. Over the years, I have developed a profile that combines rigorous scientific training with long-term applied research and system-level engineering experience. I am particularly motivated by research environments such as FLOTTANT that integrate advanced mathematical formulation, dynamic system analysis, and experimental validation under real operating constraints. I believe this dual academic and industrial trajectory positions me well to contribute effectively to the scientific and technological objectives of your project.

My doctoral research focused on the nonlinear modeling and optimization of standalone photovoltaic systems. I developed multi-level mathematical representations of photovoltaic generators and electrochemical storage systems, including parameter identification under real environmental variability, optimization-based control strategies, and state estimation using Extended Kalman Filtering for state-of-charge monitoring. In addition, I designed MPPT algorithms and implemented a dual-axis solar tracking platform to enhance system efficiency under dynamic irradiance and temperature conditions. These contributions were not confined to theoretical formulation: the models were translated into embedded architectures (C++/Python), integrated with real-time multi-sensor acquisition chains, and validated experimentally through controlled environmental testing. This integration of analytical rigor, control-oriented modeling, and full experimental deployment aligns strongly with the methodological requirements of large-scale European research projects.

Following my PhD, I pursued a decade of applied R&D leadership as Co-founder and R&D Lead of NOVEL-TI. During this period, I supervised the end-to-end development of intelligent hardware–software systems, including photovoltaic-powered platforms, embedded multi-sensor architectures, AI-enabled IoT infrastructures, and cloud-based supervision dashboards. This experience strengthened my ability to move from analytical modeling to deployable systems, ensuring robustness, scalability, and compliance with industrial standards. I also coordinated multidisciplinary teams and collaborated with European industrial partners, reinforcing my capacity to operate in international research environments.

The FLOTTANT project's focus on dynamic system modeling, data-driven analysis, and performance optimization resonates strongly with my background. I am comfortable working across abstraction levels — from differential equation modeling and state-space analysis to algorithm implementation and validation on physical platforms. My expertise in nonlinear estimation, optimization-based control, and experimental system validation enables me to contribute to the development of reliable models capable of capturing complex and time-varying interactions.

I am particularly motivated to contribute to the scientific structuring of the FLOTTANT project, including methodological formalization, reproducibility of experimental results, and dissemination through peer-reviewed publications and European research networks. My objective is to reinforce the scientific robustness of the project while ensuring that modeling developments remain directly connected to measurable system performance and technological transferability.

I am confident that my expertise in nonlinear system modeling, estimation theory, optimization-based control, embedded implementation, and industrial deployment can provide immediate added value to the FLOTTANT consortium. Beyond research contributions, I am eager to actively participate in collaborative proposal development, publication efforts, and interdisciplinary exchanges within the European framework.

Thank you for considering my application. I would welcome the opportunity to further discuss how my profile aligns with the scientific challenges and ambitions of the FLOTTANT project.

Yours faithfully,

Ferdaous Masmoudi

Attached: curriculum vitæ

Comité de recrutement du projet FLOTTANT

Sfax, 23 février 2026

*Université de Sfax
École Nationale d'Ingénieurs de Sfax (ENIS)
Sfax, Tunisie*

Madame, Monsieur,

Je me permets de vous adresser ma candidature au poste postdoctoral proposé dans le cadre du projet FLOTTANT. Titulaire d'un doctorat en Génie Électrique, spécialisé en systèmes d'énergies renouvelables, modélisation mathématique non linéaire, commande optimale et estimation d'état, j'ai développé au fil des années un profil combinant rigueur scientifique, recherche appliquée et expertise en ingénierie de systèmes complexes. Les environnements de recherche intégrant la formulation mathématique avancée, l'analyse dynamique des systèmes et la validation expérimentale sous contraintes réelles — tels que le projet FLOTTANT — correspondent pleinement à mon orientation scientifique et professionnelle. Cette double trajectoire académique et industrielle me permet d'envisager une contribution efficace aux objectifs scientifiques et technologiques du projet.

Mes travaux doctoraux ont porté sur la modélisation et l'optimisation de systèmes photovoltaïques autonomes, considérés comme des systèmes dynamiques non linéaires soumis à des contraintes physiques réelles. J'ai développé des modèles multi-niveaux intégrant l'identification paramétrique sous variabilité environnementale, l'estimation d'état par le filtre de Kalman étendu et des stratégies de commande MPPT optimisées. Les modèles ont été implémentés sur des architectures embarquées (C++/Python), intégrés à des chaînes d'acquisition multi-capteurs et validés expérimentalement sur une plateforme physique. Cette articulation entre formulation mathématique et validation en conditions réelles correspond pleinement aux exigences méthodologiques d'un projet structurant tel que FLOTTANT.

À l'issue de ma thèse, j'ai exercé pendant près de dix ans des fonctions de direction en R&D en tant que cofondatrice et responsable R&D de NOVEL-TI. J'y ai supervisé le développement complet de systèmes matériels-logiciels intelligents : plateformes énergétiques autonomes, architectures embarquées multi-capteurs, infrastructures IoT intégrant des modules d'intelligence artificielle et tableaux de supervision cloud. Cette expérience m'a permis de transformer des modèles analytiques en systèmes robustes et industrialisables, tout en garantissant la cohérence méthodologique et la traçabilité expérimentale.

L'orientation scientifique du projet FLOTTANT — modélisation dynamique avancée, analyse sous contraintes réelles, optimisation des performances et confrontation expérimentale — s'inscrit directement dans la continuité de mon parcours. Je suis particulièrement à l'aise dans le travail multi-niveaux allant de la formulation d'équations différentielles et de modèles d'état jusqu'à l'implémentation algorithmique et la validation sur plateformes physiques.

Au-delà de mes compétences techniques, j'accorde une importance particulière à la structuration collective des travaux de recherche. Mon expérience en coordination d'équipes pluridisciplinaires m'a permis de développer une approche collaborative fondée sur la clarté méthodologique, la documentation rigoureuse et la reproductibilité

des résultats. Dans un contexte transfrontalier tel que celui du programme Interreg Italie–Tunisie, je suis pleinement consciente des enjeux de coordination scientifique, d’harmonisation des protocoles expérimentaux et de valorisation commune des résultats. Je suis motivée à contribuer activement à cette dynamique collaborative et à renforcer la visibilité scientifique du projet à l’échelle européenne.

Je souhaite ainsi contribuer activement à la consolidation scientifique du projet, notamment en matière de formalisation méthodologique, de reproductibilité expérimentale et de valorisation académique à travers des publications à comité de lecture. Mon objectif est de renforcer la robustesse scientifique du projet tout en assurant un lien constant entre le développement théorique et la performance mesurable.

Convaincue que mon expertise en modélisation non linéaire, estimation avancée, commande optimisée et déploiement expérimental constitue une valeur ajoutée immédiate pour le consortium FLOTTANT, je reste à votre disposition pour tout entretien permettant de détailler l’adéquation de mon profil aux enjeux scientifiques et stratégiques du projet.

Je vous prie d’agréer, Madame, Monsieur, l’expression de ma considération distinguée.

Ferdaous Masmoudi

*encl: Curriculum vitæ,
PhD Thesis Summary*

Title: *Optimization of Autonomous Photovoltaic Systems*

My doctoral research focuses on the optimization and advanced modeling of autonomous photovoltaic systems considered as nonlinear dynamical systems subject to physical, environmental, and energy constraints. The developed approach relies on a coherent articulation between mathematical formulation, dynamic analysis, and experimental validation under real operating conditions.

1. Mathematical Modeling of the Photovoltaic Generator

Electrical models of photovoltaic cells (ideal, single-diode, double-diode, and triple-diode models) were reformulated as parameterized nonlinear systems of equations depending on thermal and irradiance variables. Internal parameter identification was treated as a constrained optimization problem incorporating sensitivity analysis and local stability assessment around the maximum power point. The models were experimentally validated to ensure consistency between theoretical formulation and real physical behavior.

2. Structural and Energy Optimization

Partial shading, electrical mismatch, and thermal losses were analyzed as dynamic perturbations affecting the stability of the operating point. Series/parallel module interconnections were studied as a structural optimization problem aimed at maximizing an energy-based performance criterion under physical constraints. A dual-axis solar tracking system was designed to introduce coupling between mechanical dynamics and the energy model, ensuring convergence toward an optimal configuration under real environmental constraints.

3. State-Space Modeling and Control of the Generation–Conversion–Storage System

The DC-DC converter was modeled in state-space representation, integrating real conduction and switching losses. The MPPT strategy was analyzed in terms of convergence and local stability. The experimentally identified battery model was incorporated into a state observation framework to optimize energy management under real operating constraints.

The main scientific contribution lies in the coherent formulation of an autonomous energy system as a global problem of dynamic modeling and optimization under real physical constraints. The overall methodology combines:

- explicit mathematical formulation of nonlinear dynamical systems,
- constrained parametric optimization,
- local stability analysis,
- experimental validation under real operating conditions.

This approach naturally aligns with the scientific objectives of the FLOTTANT project by articulating advanced mathematical formulation, dynamic analysis under real constraints, and experimental confrontation.

Keywords: Autonomous photovoltaic system, nonlinear modeling, parameter identification, multi-diode models, constrained optimization, local stability analysis, power electronics, DC-DC Boost converter, MPPT control, dual-axis solar tracking, state-space modeling, energy management, experimental validation.

Titre : *Optimisation de systèmes photovoltaïques autonomes*

Mes travaux de doctorat portent sur l'optimisation et la modélisation avancée de systèmes photovoltaïques autonomes considérés comme des systèmes dynamiques non linéaires soumis à des contraintes physiques, environnementales et énergétiques. L'approche développée repose sur une articulation cohérente entre formulation mathématique, analyse dynamique et validation expérimentale en conditions réelles.

1. Modélisation mathématique du générateur photovoltaïque

Les modèles électriques de cellules solaires (idéal, une diode, deux diodes, trois diodes) ont été reformulés comme des systèmes d'équations non linéaires paramétrées dépendant de variables thermiques et radiatives.

L'identification des paramètres internes a été traitée comme un problème d'optimisation sous contraintes intégrant analyse de sensibilité et étude de stabilité locale autour du point de puissance maximale. Cette démarche permet d'évaluer la robustesse des solutions vis-à-vis des perturbations externes (température, éclairage). Les modèles ont été validés expérimentalement afin d'assurer la cohérence entre formalisation théorique et comportement physique réel.

2. Optimisation structurelle et énergétique

Les phénomènes d'ombrage partiel, de maladaptation électrique et de pertes thermiques ont été étudiés comme des perturbations dynamiques affectant la stabilité du point de fonctionnement. L'association série/parallèle des modules a été analysée comme un problème d'optimisation structurelle. Un suiveur solaire à deux axes a été conçu afin d'introduire un couplage entre dynamique mécanique et modèle énergétique, garantissant la convergence vers une configuration optimale sous contraintes environnementales réelles. Cette approche met en évidence l'interaction entre systèmes multi-physiques et contraintes environnementales réelles.

3. Modélisation d'état et commande du système génération–conversion–stockage

Le convertisseur DC-DC a été modélisé dans l'espace d'état en intégrant les pertes réelles. La stratégie MPPT a été analysée en termes de convergence et de stabilité locale. Le modèle de batterie, identifié expérimentalement, a été intégré dans une logique d'observation d'état pour optimiser la gestion énergétique.

L'apport scientifique principal réside dans la formulation cohérente d'un système énergétique autonome comme un problème global de modélisation dynamique et d'optimisation sous contraintes physiques réelles. L'ensemble repose sur une méthodologie combinant :

- formulation mathématique explicite de systèmes dynamiques non linéaires,
- optimisation paramétrique sous contraintes physiques,
- analyse de stabilité locale,
- validation expérimentale en environnement réel.

Cette approche s'inscrit naturellement dans les thématiques du projet FLOTTANT, en articulant formulation mathématique avancée, analyse dynamique sous contraintes réelles et confrontation expérimentale.

Mots-clés : Système photovoltaïque autonome, modélisation non linéaire, identification paramétrique, modèles multi-diodes, optimisation sous contraintes, analyse de stabilité locale, électronique de puissance, convertisseur DC-DC Boost, commande MPPT, suiveur solaire à deux axes, modélisation d'état, gestion énergétique, validation expérimentale.